Asmaa Abdul-Amin

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Summary

Software Engineer with 2+ years of experience building production web applications, REST APIs, backend systems, and automation tools supporting NASA mission platforms at The Johns Hopkins University Applied Physics Laboratory. Experienced across the full software development lifecycle using Python, PHP, JavaScript, SQL, and cloud-native development practices. **Previously held an Interim Secret clearance** (currently inactive and eligible for reactivation through a sponsoring employer).

Skills

- Programming Languages: **Python, PHP, JavaScript, React, SQL, HTML, CSS**
- Frameworks & Technologies: **Bootstrap, MySQL, REST APIs, DataTables, Docker, Linux, Git, ETL Pipelines**
- Libraries & Tools: **PyTorch, TensorFlow, Hugging Face, OpenAI API, Kubernetes, Terraform, Jupyter, Atlassian, Bitbucket**

Experience

Johns Hopkins Applied Physics Laboratory

Software Engineering Intern

Space Exploration Sector (SOF-2) | May 2024 – May 2026

- Developed and maintained production web applications supporting **10+ active NASA and APL missions**, improving content management and engineering workflows for mission scientists and technical teams.
- Designed and implemented **database-driven administrative portals** and **public-facing web applications** using **PHP**, JavaScript, MySQL, Bootstrap, and DataTables, across more than a dozen mission platforms enabling mission teams to manage content without code changes.
- Designed and implemented backend REST APIs and database services powering event management, newsletters, FAQ systems, and mission repositories used across multiple mission websites.
- Modernized legacy web applications by replacing paginated interfaces with **searchable and responsive components**, improving usability and simplifying content discovery for mission stakeholders.
- **Planned, estimated, and implemented major features** for the Ice Giants mission website, including content migration, workshop archives, and administrative tooling.
- Collaborated directly with scientists, project managers, and technical staff to **translate requirements into deployed software solutions** supporting active NASA mission platforms.
- Developed Python **automated utilities** and **database export tools** that generate **JSON feeds** from mission databases to support downstream analysis and **integration with internal engineering systems.**
- Owned features from requirements gathering and implementation through testing, deployment, maintenance, and iterative enhancements across **multiple concurrent projects**.

Undergraduate Research Intern

Research & Exploratory Development Department (R1N) | Mar 2023 – May 2024

- Engineered and **deployed multiple webpages for the NASA Dragonfly** internal portal, streamlining access to critical mission resources and improving mission collaboration efficiency.
- **Developed machine learning models analyzing COVID-19 policies** vs. mortality, improving interpretability of public health data.

Projects

Parker Solar Probe - Developed and maintained the Parker-Kit Celebration webpage.
Ice Giants - Designed and implemented the Ice Giants research website.
EZIE - Rebuilt the EZIE website from ReactJS to PHP.
LOGIC - Developed a database-backed terminology management system for standardized lunar infrastructure definitions.
LSIC - Developed JSON APIs and optimized database-backed newsletter management.
BLKPVNTHR.OS – Engineering an AI-powered personal operating system featuring specialized autonomous agents and workflows.

Education

B.S. Data Science – University of Maryland Global Campus (Expected Dec 2028)

A.A. Computer Science – Montgomery College (Expected Dec 2026)

Certifications:

The Johns Hopkins University Applied Physics Laboratory – Python Programming II (2025)
Univ. of Chicago – *Quantum Computing Systems Design I* (2024)
IBM – *Machine Learning with Python* (2023)
Harvard – *CS50 Intro to Computer Science* (2022)